A Restriction Theorem for Springer Fibers

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Abstract

Let G be a connected complex reductive group, $\mathcal{B}$ its flag variety, and $\mathcal{B}_e$ the Springer fiber of a nilpotent element $e \in \operatorname{Lie} G$. Let S be the torus associated with an $\mathfrak{sl}_2$-triple containing e. We prove that the image of the restriction map on complex cohomology $H^*(\mathcal{B}^S) \to H^*(\mathcal{B}_e^S)$ between the S-fixed point loci is the subalgebra of invariants under the component group $A(e)$ of the centralizer of e in G. This proves a conjecture of Hoang, Krylov, and Matvieievskyi. The argument uses global invariant cycles. As an application, we give another proof of the known surjectivity for general linear groups.

1 Introduction

Let G be a connected complex reductive group with Lie algebra $\mathfrak{g}$ and flag variety $\mathcal{B}$ of Borel subalgebras of $\mathfrak{g}$. Write Ad for the adjoint action of G on $\mathfrak{g}$ and its induced action on $\mathcal{B}$. For $x \in \mathfrak{g}$, let $Z_G(x) = \{g \in G \mid \mathrm{Ad}(g)x = x\}$ be its centralizer in G.

Fix a nilpotent element $e \in \mathfrak{g}$, an $\mathfrak{sl}_2$-triple (e, h, f), and the associated cocharacter $\lambda\colon \mathbb{C}^\times \to G$, with $d\lambda(1) = h$. Set

$$S = \lambda(\mathbb{C}^\times), \qquad A(e) = Z_G(e)/Z_G(e)^\circ,$$

where $Z_G(e)^\circ$ is the identity component of $Z_G(e)$. For $e = 0$, we take the trivial triple and cocharacter. Since $\mathrm{Ad}(\lambda(t))e = t^2 e$, the torus S preserves the Springer fiber

$$\mathcal{B}_e = \{\mathfrak{b} \in \mathcal{B} \mid e \in \mathfrak{b}\}.$$

Throughout the paper, $H^*(X)$ denotes singular cohomology with coefficients in $\mathbb{C}$. For a group H acting on a variety X, write X^H for its fixed point locus. A superscript H on a cohomology group denotes the subspace of H-invariant classes. Let $i\colon \mathcal{B}_e^S \hookrightarrow \mathcal{B}^S$ be the inclusion. The centralizer of the triple acts on $\mathcal{B}_e^S$, and its action on cohomology factors through $A(e)$. We describe this action in Section 2.

The following theorem proves [HKM24, Conjecture 10.10]. Some classical cases were established in [HKM24, Lemma 10.9].

Theorem 1.1. *The pullback map $i^*\colon H^*(\mathcal{B}^S) \to H^*(\mathcal{B}_e^S)$ satisfies*

$$\mathrm{Im}(i^*) = H^*(\mathcal{B}_e^S)^{A(e)}.$$

In particular, i^* is surjective if and only if $A(e)$ acts trivially on $H^*(\mathcal{B}_e^S)$. Since each connected component of $\mathcal{B}^S$ is a flag variety, the invariant algebra is generated in degrees zero and two.

For $G = \mathrm{GL}_n$, the group of invertible $n \times n$ complex matrices with $n \geq 1$, the component group $A(e)$ is trivial, so Theorem 1.1 recovers the known surjectivity of i^* (Corollary 3.5). This also follows from the classical restriction theorem for ordinary Springer fibers [DCP81, Tan82], by equivariant formality and localization. Outside type A, the image of $H^*(\mathcal{B}) \to H^*(\mathcal{B}_e)$ can be a proper subalgebra of $H^*(\mathcal{B}_e)^{A(e)}$ [HKM24, Warning 10.19].

We apply Deligne's global invariant cycle theorem [Del71] to the incidence family of De Concini, Lusztig, and Procesi [DLP88, Sections 2.1–2.4 and 3.7]. The intersections of $\mathcal{B}_e^S$ with the connected components of $\mathcal{B}^S$ also occur in the decomposition of nilpotent Hessenberg cohomology as $A(e)$-modules studied by Precup and Sommers [PS25, Section 3.2 and Corollary 3.6].

At positive admissible slopes, Oblomkov and Yun identify the image of ambient flag cohomology with the invariants under the stabilizer component group and braid-group monodromy for Hessenberg varieties arising from regular semisimple homogeneous affine Springer fibers [OY16, Theorem 5.5.1(1)]. Their proof uses the decomposition theorem and contraction of the incidence family. We apply global invariant cycles over the open orbit $Z_G(h) \cdot e$.

2 Fixed flags and the incidence family

All Springer fibers and fixed point varieties are endowed with their reduced scheme structures.

For a group element or subgroup, $Z_G(-)$ denotes its centralizer under conjugation. For several arguments, it denotes their common centralizer. We write H° for the identity component of an algebraic group H.

Let a complex algebraic group H act on a variety X, and write $m_a \colon X \to X$ for the automorphism induced by $a \in H$. We let a act on cohomology by $m_{a^{-1}}^*$. The complex Lie group $H^\circ(\mathbb{C})$ is path connected. For $a \in H^\circ$, a path from 1 to a induces a homotopy between id_X and m_a. Hence H° acts trivially on $H^*(X)$, and the cohomological action of H factors through H/H°.

For any connected component F of $\mathcal{B}^S$, put $F_e = F \cap \mathcal{B}_e^S$ and write $i_F \colon F_e \hookrightarrow F$ for the inclusion.

Retain the triple (e, h, f) and cocharacter λ from the introduction. For $j \in \mathbb{Z}$, set

$$\mathfrak{g}(j) = \{x \in \mathfrak{g} \mid [h, x] = jx\}, \qquad L = Z_G(S) = Z_G(h), \qquad Z_e = Z_G(e, h, f).$$

Here $Z_G(S)$ consists of the elements of G commuting with every element of S, and $Z_G(e, h, f) = Z_G(e) \cap Z_G(h) \cap Z_G(f)$. The Levi subgroup $L \subset G$ has Lie algebra $\mathfrak{g}(0)$ [Hum75]. For $e = 0$, these conventions give $L = Z_e = G$ and $\mathfrak{g}(2) = 0$.

Lemma 2.1. *Let M be a finite-dimensional $\mathfrak{sl}_2$-module and write $M_j = \{v \in M \mid hv = jv\}$ for $j \in \mathbb{Z}$. Then*

$$e(M_0) = M_2, \qquad \ker(e) \cap M_{-2} = 0,$$

and every weight of $\ker(e)$ is nonnegative.

Proof. Decompose M into irreducible modules. On each summand, the kernel of e is the highest-weight line. Moreover, e maps the weight-zero space onto the weight-two space whenever the latter is nonzero. □

Lemma 2.2. *The orbit $L \cdot e$ is open in $\mathfrak{g}(2)$, and the centralizer $Z_L(e)$ of e in L equals Z_e.*

Proof. By Lemma 2.1, the differential at 1 of the orbit map $L \to \mathfrak{g}(2)$, $g \mapsto \mathrm{Ad}(g)e$, is the surjection $\mathfrak{g}(0) \to \mathfrak{g}(2)$, $a \mapsto [a, e]$. Hence $L \cdot e$ is open in $\mathfrak{g}(2)$.

Writing $\mathrm{ad}(e) = [e, -]$, we have, for $g \in Z_L(e)$,

$$\mathrm{Ad}(g)f - f \in \mathfrak{g}(-2) \cap \ker(\mathrm{ad}\, e) = 0.$$

Thus g centralizes the triple, and the reverse inclusion is immediate. □

Proposition 2.3. *The inclusion $Z_e \hookrightarrow Z_G(e)$ induces an isomorphism*

$$Z_e/Z_e^\circ \xrightarrow{\sim} A(e).$$

Proof. Choose a faithful representation $G \hookrightarrow \mathrm{GL}(V)$ on a finite-dimensional complex vector space V, where $\mathrm{GL}(V)$ is its group of linear automorphisms. Identify $\mathfrak{g}$ with its image in the endomorphism algebra $\mathrm{End}(V)$. For $g \in Z_G(e)$, both g and g^{-1} lie in the kernel of $[e, -]$ on $\mathrm{End}(V)$. By Lemma 2.1, their weight decompositions under conjugation by λ have no negative-weight components. Thus the two maps

$$t \longmapsto \lambda(t)g\lambda(t)^{-1}, \qquad t \longmapsto \lambda(t)g^{-1}\lambda(t)^{-1}$$

extend regularly to $t = 0$. The two limiting matrices are inverse to one another and belong to the closed subgroup $G \subset \mathrm{GL}(V)$. Denote the first limit by $r(g)$. Since $Z_G(e)$ is closed and stable under conjugation by $\lambda(t)$, we have $r(g) \in Z_G(e)$. The matrix $r(g)$ has weight zero, so it centralizes S. By Lemma 2.2, we obtain a retraction

$$r\colon Z_G(e) \longrightarrow Z_e, \qquad r(g) = \lim_{t\to 0} \lambda(t)g\lambda(t)^{-1}.$$

The map is algebraic, being the restriction of the weight-zero projection on $\mathrm{End}(V)$, and fixes Z_e pointwise.

The extended map from the affine line $\mathbb{A}^1$ to $Z_G(e)$ takes 1 to g and 0 to $r(g)$, so these elements lie in the same connected component. Thus every connected component of $Z_G(e)$ meets Z_e. Conversely, if $c \in Z_e \cap Z_G(e)^\circ$, then $c = r(c)$ belongs to the connected set $r(Z_G(e)^\circ)$ containing the identity. Hence

$$Z_e \cap Z_G(e)^\circ = Z_e^\circ,$$

proving the assertion. □

The triple centralizer Z_e preserves $\mathcal{B}_e^S$, and its cohomological action factors through $Z_e/Z_e^\circ \simeq A(e)$. This is the action used in Theorem 1.1. The inclusion $\mathcal{B}_e^S \hookrightarrow \mathcal{B}_e$ is Z_e-equivariant, so restriction is compatible with the usual $A(e)$-action on $H^*(\mathcal{B}_e)$.

Lemma 2.4. *Every connected component of $\mathcal{B}^S$ is an orbit of the Levi subgroup L. If B_0 is the Borel subgroup corresponding to $\mathfrak{b}_0 \in \mathcal{B}^S$, then*

$$L \cdot \mathfrak{b}_0 \simeq L/(L \cap B_0),$$

where $L \cap B_0$ is a Borel subgroup of L.

Proof. Fix a maximal torus $T \subset G$ containing S. A Borel subgroup normalized by S contains S, since Borel subgroups are self-normalizing. Conjugacy of maximal tori in L shows that every such subgroup is L-conjugate to a Borel subgroup containing T. There are only finitely many Borel subgroups containing T, so $\mathcal{B}^S$ has finitely many L-orbits. If $T \subset B_0$, then $L \cap B_0$ is the Borel subgroup of L determined by the corresponding positive roots [Hum75]. Each orbit is therefore a projective flag variety. The finitely many orbits are closed and connected, hence are precisely the connected components. □

Fix a connected component $F \subset \mathcal{B}^S$ and a point $\mathfrak{b}_0 \in F$, and let B_0 be the corresponding Borel subgroup. Set $B_L = L \cap B_0$, a Borel subgroup of L. Consider

$$\mathcal{E}_F = \{(x, \mathfrak{b}) \in \mathfrak{g}(2) \times F \mid x \in \mathfrak{b}\}, \qquad \pi_F \colon \mathcal{E}_F \longrightarrow \mathfrak{g}(2), \quad (x, \mathfrak{b}) \longmapsto x.$$

This is the incidence construction of [DLP88, Sections 2.1–2.4 and 3.7].

Proposition 2.5. *The incidence variety $\mathcal{E}_F$ is a vector subbundle of $\mathfrak{g}(2) \times F \to F$, and π_F is projective. Let $\mathcal{O}_F$ denote the trivial line bundle on F, and let $\mathbb{P}_F$ parametrize lines in a vector bundle over F. The projective bundle*

$$\overline{\mathcal{E}}_F = \mathbb{P}_F(\mathcal{E}_F \oplus \mathcal{O}_F),$$

is a smooth projective compactification of $\mathcal{E}_F$. If $F_e \neq \varnothing$, the restriction

$$\pi_F^{-1}(L \cdot e) \longrightarrow L \cdot e$$

is smooth and surjective, and $\pi_F^{-1}(L \cdot e)$ is dense in $\overline{\mathcal{E}}_F$. In particular, F_e is smooth and projective.

Proof. There is an isomorphism over $F = L/B_L$,

$$L \times^{B_L} (\mathfrak{g}(2) \cap \mathfrak{b}_0) \xrightarrow{\sim} \mathcal{E}_F, \qquad [g, x] \longmapsto (\mathrm{Ad}(g)x, \mathrm{Ad}(g)\mathfrak{b}_0).$$

Here $(g, x) \sim (gb, \mathrm{Ad}(b^{-1})x)$ for $b \in B_L$. Zariski local sections of $L \to F$ show that this map is an isomorphism. The B_L-equivariant inclusion $\mathfrak{g}(2) \cap \mathfrak{b}_0 \hookrightarrow \mathfrak{g}(2)$ identifies $\mathcal{E}_F$ with a vector subbundle of

$$L \times^{B_L} \mathfrak{g}(2) \simeq \mathfrak{g}(2) \times F, \qquad [g, x] \longmapsto (\mathrm{Ad}(g)x, gB_L).$$

The total space of $\mathcal{E}_F$ is smooth and irreducible. Since $\mathcal{E}_F$ is closed in $\mathfrak{g}(2) \times F$ and F is projective, π_F is projective.

Equivariance implies surjectivity over $L \cdot e$ whenever F_e is nonempty. At a point above $x \in L \cdot e$, the image of the differential of π_F contains the image of the infinitesimal orbit map

$$\mathfrak{g}(0) \longrightarrow \mathfrak{g}(2), \qquad a \longmapsto [a, x].$$

By Lemma 2.2, this map is surjective. Since $\mathcal{E}_F$ and $\mathfrak{g}(2)$ are smooth, π_F is smooth over $L \cdot e$. Its fiber over e is therefore reduced and is identified with F_e by projection to F.

The projective bundle $\overline{\mathcal{E}}_F$ is smooth and projective. It contains $\mathcal{E}_F$ as the open subset where the $\mathcal{O}_F$-coordinate is nonzero. The inverse image of $L \cdot e$ is nonempty and open in the irreducible variety $\mathcal{E}_F$, hence dense in $\overline{\mathcal{E}}_F$. □

3 The restriction theorem

For a complex variety X, let $\pi_0(X)$ be its set of connected components and $\pi_1(X, x)$ its topological fundamental group based at $x \in X$.

Lemma 3.1. *If $F_e \neq \varnothing$, then for every $k \geq 0$, the monodromy representation of $\pi_F^{-1}(L \cdot e) \to L \cdot e$ on $H^k(F_e)$ has the same image as the $A(e)$-action. Thus*

$$H^k(F_e)^{\pi_1(L \cdot e, e)} = H^k(F_e)^{A(e)}.$$

Proof. By Lemma 2.2, the family is the associated bundle

$$L \times^{Z_e} F_e \xrightarrow{\sim} \pi_F^{-1}(L \cdot e), \qquad [g, \mathfrak{b}] \longmapsto (\mathrm{Ad}(g)e, \mathrm{Ad}(g)\mathfrak{b})$$

over L/Z_e, with $(g, \mathfrak{b}) \sim (gc, \mathrm{Ad}(c^{-1})\mathfrak{b})$ for $c \in Z_e$. The analytic principal Z_e-bundle $L \to L/Z_e$ has local sections. A loop based at e lifts to a path in L from 1 to an element $c \in Z_e$. Its transport on cohomology is $m_{c^{-1}}^*$, where $m_c \colon F_e \to F_e$ is the automorphism $\mathfrak{b} \mapsto \mathrm{Ad}(c)\mathfrak{b}$. Two lifts of the same loop starting at 1 have endpoints in the same connected component of Z_e, so they induce the same cohomology operator. Conversely, since the Levi subgroup L is connected, every $c \in Z_e$ can be joined to 1 by a path in L whose projection to L/Z_e is a loop. The monodromy image is therefore the image of $Z_e/Z_e^\circ \simeq A(e)$. □

Theorem 3.2 (Deligne)**.** *Let $p \colon \mathcal{X} \to Y$ be a smooth projective surjective morphism of complex varieties with Y smooth and connected, and let $\overline{\mathcal{X}}$ be a smooth projective compactification of $\mathcal{X}$. For $y \in Y$, write $\mathcal{X}_y = p^{-1}(y)$. Then, for $k \geq 0$,*

$$\mathrm{Im}\big(H^k(\overline{\mathcal{X}}) \longrightarrow H^k(\mathcal{X}_y)\big) = H^k(\mathcal{X}_y)^{\pi_1(Y, y)}.$$

This is [Del71, Théorème 4.1.1(ii)], after extension of scalars from $\mathbb{Q}$ to $\mathbb{C}$. The fibers need not be connected, and p need not extend to $\overline{\mathcal{X}}$.

Proof of Theorem 1.1. Fix $k \geq 0$ and a connected component $F \subset \mathcal{B}^S$ with $F_e \neq \varnothing$. Since the triple centralizer Z_e lies in the connected Levi subgroup L, it acts trivially on $H^*(F)$. Equivariance of i_F implies

$$\mathrm{Im}(i_F^*) \subseteq H^*(F_e)^{A(e)}.$$

Let $\alpha \in H^k(F_e)^{A(e)}$. By Lemma 3.1, it is invariant under the monodromy of the smooth projective family over $L \cdot e$. By Proposition 2.5, $\overline{\mathcal{E}}_F$ is a smooth projective compactification

of $\pi_F^{-1}(L \cdot e)$. Write $j_F \colon F_e \hookrightarrow \overline{\mathcal{E}}_F$ for the embedding $\mathfrak{b} \mapsto (e, \mathfrak{b})$. By Theorem 3.2, there exists $\beta \in H^k(\overline{\mathcal{E}}_F)$ such that $j_F^* \beta = \alpha$.

Let $\sigma_F \colon F \to \mathcal{E}_F \subset \overline{\mathcal{E}}_F$ be the zero section and put $\gamma = \sigma_F^* \beta$. The maps $\sigma_F \circ i_F$ and j_F are homotopic through

$$F_e \times [0,1] \longrightarrow \mathcal{E}_F, \qquad (\mathfrak{b}, t) \longmapsto (te, \mathfrak{b}).$$

This homotopy is well defined because $e \in \mathfrak{b}$ for $\mathfrak{b} \in F_e$. Hence

$$i_F^* \gamma = (\sigma_F \circ i_F)^* \beta = j_F^* \beta = \alpha.$$

We have proved

$$\operatorname{Im}\bigl(i_F^* \colon H^k(F) \longrightarrow H^k(F_e)\bigr) = H^k(F_e)^{A(e)}. \tag{1}$$

If $F_e = \varnothing$, both sides of (1) vanish. Since $Z_e \subset L$, it preserves each F and F_e. Taking the direct sum of (1) over the connected components F of $\mathcal{B}^S$ proves the assertion in every degree. Restriction preserves cup products, so the equality respects the algebra structures. □

The action of Z_e on $\pi_0(\mathcal{B}_e^S)$ factors through $A(e)$, since Z_e° is connected.

Corollary 3.3. *For every connected component F of $\mathcal{B}^S$ with $F_e \neq \varnothing$, the group $A(e)$ acts transitively on the connected components of F_e. In particular, F_e is connected if $A(e)$ is trivial.*

Proof. By (1), we have $H^0(F_e)^{A(e)} = \mathbb{C}$. The left-hand side identifies with the algebra of functions on $\pi_0(F_e)/A(e)$, so this orbit set has one element. □

This transitivity is also proved in [DLP88, Lemma 2.2(iii)].

Corollary 3.4. *The algebra $H^*(\mathcal{B}_e^S)^{A(e)}$ is generated by its degree-zero and degree-two parts. In particular,*

$$H^{2j+1}(\mathcal{B}_e^S)^{A(e)} = 0 \qquad (j \geq 0).$$

Proof. For each connected component F of $\mathcal{B}^S$, choose a Borel subgroup $B_L \subset L$ so that $F \simeq L/B_L$. Let $X^*(B_L)$ be its character group and, for $\chi \in X^*(B_L)$, let $\mathbb{C}_\chi$ be the one-dimensional B_L-module with character χ. Writing Sym for the symmetric algebra over $\mathbb{C}$ and c_1 for the first Chern class, Borel's characteristic homomorphism

$$\operatorname{Sym}\bigl(X^*(B_L) \otimes_{\mathbb{Z}} \mathbb{C}\bigr) \longrightarrow H^*(F), \qquad \chi \longmapsto c_1(L \times^{B_L} \mathbb{C}_\chi),$$

is surjective [Bor53]. The generators $X^*(B_L) \otimes_{\mathbb{Z}} \mathbb{C}$ have degree two. This applies to the connected reductive group L, since a finite central cover induces an isomorphism of flag varieties and an isomorphism of character spaces after tensoring with $\mathbb{C}$. Thus $H^*(\mathcal{B}^S)$ is generated by these Chern classes and the idempotents corresponding to its connected components. Apply Theorem 1.1. □

The stronger vanishing $H^{2j+1}(\mathcal{B}_e^S) = 0$ is proved in [DLP88, Theorem 3.9]. For $e \neq 0$, apply their theorem to $s = \lambda(t)$ with t not a root of unity. The fixed point locus $\mathcal{B}_e^s$ of s then equals $\mathcal{B}_e^S$. The case $e = 0$ is the usual vanishing for $\mathcal{B}$.

Sending a connected component of $\mathcal{B}_e^S$ to the connected component of $\mathcal{B}^S$ containing it induces, by Corollary 3.3, a bijection

$$\pi_0(\mathcal{B}_e^S)/A(e) \xrightarrow{\sim} \{F \in \pi_0(\mathcal{B}^S) \mid F_e \neq \varnothing\}.$$

Corollary 3.5. *If $G = \mathrm{GL}_n$, then the pullback map*

$$i^*\colon H^*(\mathcal{B}^S) \longrightarrow H^*(\mathcal{B}^S_e)$$

is surjective.

Proof. The centralizer $Z_G(e) = Z_G(1+e)$ is connected [SS70, III, 3.22] (see also [Spr76, Section 7.7]). Hence $A(e) = 1$, and the assertion follows from Theorem 1.1. □

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